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# Large-scale effects of $H_2O$ and $O_2$ on the absorption and permeation in Nb of energetic hydrogen particles

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The influence of chemically active gases ($O_2$ and $H_2O$) on hydrogen re-emission was investigated by and permeation through a Nb membrane acted upon by 20 keV $H_2^+$ ions. Both the initial retention of ions after switching on the ion beam and the build-up of re-emission with time can be directly monitored with the new UHV apparatus by the changes of hydrogen pressure in the vacuum chamber in front of the membrane, since this pressure is determined by the incoming beam.

In the absence of $O_2$ or $H_2O$, only a small portion of the flux absorbed penetrates the membrane, and the re-emission flux virtually equal to the flux of incident ions is established following the absorption of a relatively small dose. The admittance of $O_2$ or $H_2O$ on the inlet side of the membrane continuously acted upon by the $H_2^+$-ion beam results in a dramatic increase of both the retention and the permeation of implanted hydrogen. With the pressures of $O_2$ and $H_2O$ corresponding to about 0.03 O-atoms supplied to the sample surface per impinging proton, the time to reach the steady-state re-emission sharply rises, indicating a drastic increase of the dose absorbed. The steady-state re-emission flux is not changed appreciably, while the permeation flux, though increased by orders of magnitude, remains small compared to the incident ion flux. With greater pressures of the admitted gases (more than 0.1 O-atom supplied to the surface per proton), not only the sorptional capacity is increased, but also the permeation flux reaches up to the magnitude comprising the major part of the absorbed flux (i.e. superpermeability develops), and the re-emission flux correspondingly falls.

## 1. Introduction

### *1.1. The role of the surface potential barrier*

The permeation and the retention of non-equilibrium energetic hydrogen in metals are effectively increased by the presence of potential barrier $\mathscr{E}_0$ (see fig. 1) at a metal surface [1–4]. The quantity $2\mathscr{E}_0$ represents the activation energy for dissociative binding of a thermal $H_2$ molecule impinging on the metal surface. The surface barrier also hinders the reverse process of the recombinative evolution of dissolved hydrogen, as it is clear considering the balance of fluxes at thermal equilibrium. Unlike the thermal molecules, energetic hydrogen particles (e.g. thermal atoms [3,7,8] and, especially, fast ions) readily penetrate into the metal lattice despite of the presence of the surface potential barrier, while the process of thermal evolution of the implanted hydrogen thermalized in the lattice may be greatly impeded by this barrier. So, if the evolution is governed by thermal mechanisms rather than by the stimulated (ion-induced) desorption, the permeation and the uptake of hydrogen will be the greater (and the suppression of re-emission the stronger), the higher the $\mathscr{E}_0$ barrier on the surface contacting energetic hydrogen. It is known that $\mathscr{E}_0$ equals zero for the transition metal clean surfaces [9], but $\mathscr{E}_0$ becomes much larger than $RT$ ($R$ is the molar gas constant, and $T$ is temperature), if the surface is covered by non-metal contaminant films comprising up to only about one monolayer [7,8,10,11]. Thus the evolution of dissolved hydrogen is effectively impeded for the case of "real" (i.e. insufficiently cleaned) metal surface, and this is true for the metals both exothermically and endothermically occluding hydrogen.

The notion that the metal surface chemical composition would be of prime importance for the uptake and permeation of energetic hydrogen particles has been expressed by the authors of the present paper [1,2,12–14] and by Perkins and Noda [15]. This was confirmed in

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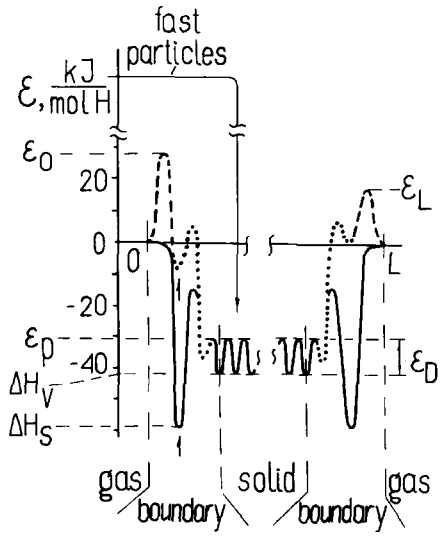



Fig. 1. The potential diagram of H–Nb system [5]. The relief for clean surface [6] is shown by solid line, that for the surface covered by non-metallic contaminant (a random example) by dashed line; the dotted line is to present a variant for the unknown part of the relief. The step $1 \rightarrow 0$ implies the association of two H atoms into an $H_2$ molecule during the process of thermal evolution.

direct experiments by these authors [8,16–19] and by other workers [20–24]. The significance of this topic for the fusion reactor technology has been specifically emphasized in several review articles [4,8,14,25–27]. The effects of surface chemistry on hydrogen retention and permeation have been observed in experiments with thermal H-atoms [7,8,12–14,16,18,19,28] and lowest energy ions (tens of eV) [20,26] that do not impair monatomic passivating films, as well as in experiments with fast ions [8,15–17,20–26,28,29] that sputter the surface. In the latter case, the non-metallic passivating film can be maintained by the contaminant segregation from the bulk metal, by the reactions with gases in the vacuum system, or by the plasma transport of matter. The data by Behrisch et al. [30] seem to be noteworthy in this respect, inferring that the non-metallic films dynamically maintained on a metal surface are able to significantly bring down the metal sputtering by hydrogen ions: in the presence of oxygen, the sputtering of iron was lowered by a factor of 10 to 15.

### *1.2. Energetic hydrogen retention*

The absorption of energetic hydrogen particles by metals has for a long time been suggested as a means of pumping and of recycling control in fusion reactors (the linear in mirror devices [31] and tokamaks [4,32], tokamak divertors, injectors for various plasma traps). A good physical background for such suggestions had been delivered by the experiments on implantation of fast hydrogen ions in metals, especially, in hydride-forming ones [33,34]. There is some evidence in these data indicating that the presence of oxygen on the surface of, say, niobium, enhances hydrogen retention in the metal. It seems, therefore, reasonable to make an attempt at maintaining a constant concentration of non-metal contaminant on the metal surface during the bombardment by fast hydrogen particles in order to substantially enlarge the metal capability to take up hydrogen before the beginning of considerable re-emission. The research of this kind may, in addition, serve to better understanding of the whole problem of hydrogen recycle in fusion devices.

### *1.3. Superpermeability*

It seems again advantageous to employ the permeation of energetic hydrogen through metal membranes for the evacuation of hydrogen isotopes from vacuum systems [1–3,14,25], fusion reactors including. With superpermeability, half or more of all hydrogen particles entering the metal lattice penetrate the membrane. Some evident benefits by the membrane pumping are: the continuity of the process (i.e. there is no dose or time limitation), and simultaneous hydrogen compression and purification (separation from helium). Superpermeability has been observed on many metals interacting with thermal hydrogen atoms (e.g. refs. [3,7,8,12–14,16–19,28,35]), and the possibility of effective pumping and compression of hydrogen isotopes has been convincingly demonstrated, for instance, by KFA Jülich researchers [35].

The feasibility of the direct membrane pumping is less evident for the case of fast hydrogen particles. Superpermeability will be obtained only at the condition that the probability of hydrogen desorption through the outlet (downstream) surface is not less than that of back re-emission. To satisfy this requirement may be not so easy. For one thing, the upstream boundary being acted upon by fast particles, the stimulated ion-induced re-emission may occur there. Even if the stimulated evolution turns out to be insignificant [27], sputtering contaminants from the upstream surface will accelerate the usual thermal evolution through destroying the $\mathscr{E}_0$ barrier. The degradation of superpermeability will inevitably result from it in the case of "natural" (i.e. covered by non-metallic adlayers) downstream

surface: will become $\mathscr{E}_L > \mathscr{E}_0$. To keep the downstream surface clean (i.e. with $\mathscr{E}_L = 0$) may be problematic by itself, but even on succeeding in this one will face several complications. Firstly, one cannot get the value of the probability of permeation at one impact, $\chi$, more than $1/2$ [1]. Secondly, the maximum density of permeation flux and the maximum membrane thickness at which superpermeability can be reached will be dramatically decreased, as compared with the membrane having non-metallic contaminant layers on the upstream surface [1,2,8,25]. Thirdly, the attainable degree of compression will be decreased [2,13]. The drift of the membrane permeability to lower values (often by many orders of magnitude) caused by the upstream surface cleaning under ion bombardment has been observed by many researchers [8,15–17,20,23,24,37,38].

Nevertheless, the authors of this paper obtained steady-state superpermeability (with $\chi > 1/2$) for $D^+$-ions of widely ranging energies [16,28] on a niobium membrane that had a "natural" outlet surface, without taking any particular steps to maintain the contamination of inlet surface. It has been suggested [16,28] that the surface passivation was being maintained spontaneously due to the surface passivation in water vapour ($(1-3)\times 10^{-6}$ Pa) that appeared in the vacuum system with switching on the $D^+$-ion beam. It has also been demonstrated that should superpermeability by niobium be destroyed through ion bombardment, it can be restored and permanently maintained by means of admitting oxygen [16], or hydrogen sulphide [8], on the inlet side of the membrane. The admittance of $O_2$ or $H_2S$ significantly increases the permeability of austenitic steel membranes to $D^+$-ions as well [8,17].

### *1.4. The research programme outlines*

As can be inferred from the above, a systematic investigation appears expedient to the role of physico-chemical factors in hydrogen re-emission, uptake and permeation in metals. Means and ways should be found to control the state of metal surfaces in the process of their interaction with fast hydrogen and helium particles. It would seem possible, among other things, to get a substantial increase in hydrogen retention and permeation by way of controlling the surface chemical composition, which could enable the effective evacuation of energetic hydrogen from fusion devices, with energies, corpuscular fluxes and power load ranges of practical importance.

With this aim in view, a new experimental apparatus has been designed and assembled at the author's laboratory. The set-up permits to continuously monitor the build-up of the re-emission and permeation fluxes that follows switching on the ion beam incident on a membrane sample whose either surface may be subject to physico-chemical treatment by the reactions with admitted gases during an experimental run.

In this paper, we present the data of our first experiments where the availabilities of the new apparatus were employed only partially. The choice of niobium as an object of the investigation was prompted by the notion that niobium, along with vanadium and tantalum, should be considered as a promising material for the sorptional or membrane pumping of energetic hydrogen particles. We had tentatively attributed the "spontaneous" superpermeability of niobium to $D^+$-ions to the presence of water vapour in our previous experiments [16,28], so it was natural to begin this research with $H_2O$ and $O_2$ purposely admitted to the membrane sample inlet side. We intentionally avoided cleaning the downstream surface of the membrane in the experiments described here, for the reasons that the "natural" state of niobium surface would seem most feasible regarding applications, as such a variant is easier to realize in practice. Nevertheless, the methods to thoroughly clean in situ the bulk and the outlet surface of a membrane sample are included in the experimental scheme, as well as the possibility for chemical protection of the outlet surface (e.g. by evaporating a palladium overlayer [8]), and we are planning the experiments on the samples cleaned to the maximum degree for the future, in order to get at a better understanding of the physico-chemical mechanisms involved in the interaction of non-equilibrium energetic hydrogen particles with metals.

## 2. Experimental methods and apparatus

The capability of our equipment and its background philosophy are roughly outlined below. In the present research, only a fraction of the experimental methods available have been actually used, but the whole set of the possibilities will be applied, it is hoped, to our further investigations.

[1] For two-layer membranes, $\chi > 1/2$ is possible even with both the inlet and outlet surfaces being clean, on the condition that the outlet surface is characterized by a higher sticking coefficient, $\alpha$, for hydrogen molecules. So, permeation with $\chi > 1/2$ was observed for a Nb–Pd membrane having the both surfaces clean [8], in accordance with that $\alpha_{Pd} > \alpha_{Nb}$ is known [36].

### 2.1. General schematic

The equipment used is shown in figs. 2 and 3. A duoplasmatron with a hollow cathode [39] is used in this set-up as an ion source. The mass-analized ion beam is directed onto a membrane sample that separates two UHV chambers pumped via calibrated diaphragms. Re-emission and permeation result in pressure increases in these chambers. The pressures are measured by a two omegatron mass-spectrometers to determine the re-emission and the permeation fluxes. The re-emission can be measured due to the fact that the incoming flux of gas on the membrane upstream side is defined by the ion beam entering the chamber rather than by the influx of diffused hydrogen. This is achieved by 3-stage pumping of the ion-beam duct, like in the experiments by McCracken and Maple [40]. When the mass-separator magnet deflects the ion beam into the upstream chamber diaphragm, the pressure there is increased ~ 10 times (e.g. $H_2^+$-beam of $4 \times 10^{14}$ $H_2^+/s$ makes the $H_2$ pressure rise to $1.3 \times 10^{-4}$ Pa). This pressure rise develops, of course, only if all the gas absorbed by the membrane sample is returned to the upstream chamber. When the sorptional, or the membrane, pumping of the beam takes place, the pressure becomes effectively lower. It is this decrease alongside of the corresponding pressure rise in the downstream chamber that is being measured during each experimental run.

### 2.2. The ion beam

The ion beam is extracted from the ion source orifice by a 20 kV voltage, is focused by three electrostatic lenses placed before and beyond the mass-analizer, passes through the electrostatic retardation system, enters, via the 1 cm diameter upstream diaphragm, the space inside the hollow cylinder "collector" (see section 2.5), and finally hits the sample surface. The potential of the collector is usually kept close to that of the retardation system end electrode, while the ion beam final deceleration occurs between the collector and the membrane (the potential of the latter being nearly +20 kV with respect to the grounded apparatus vessel when

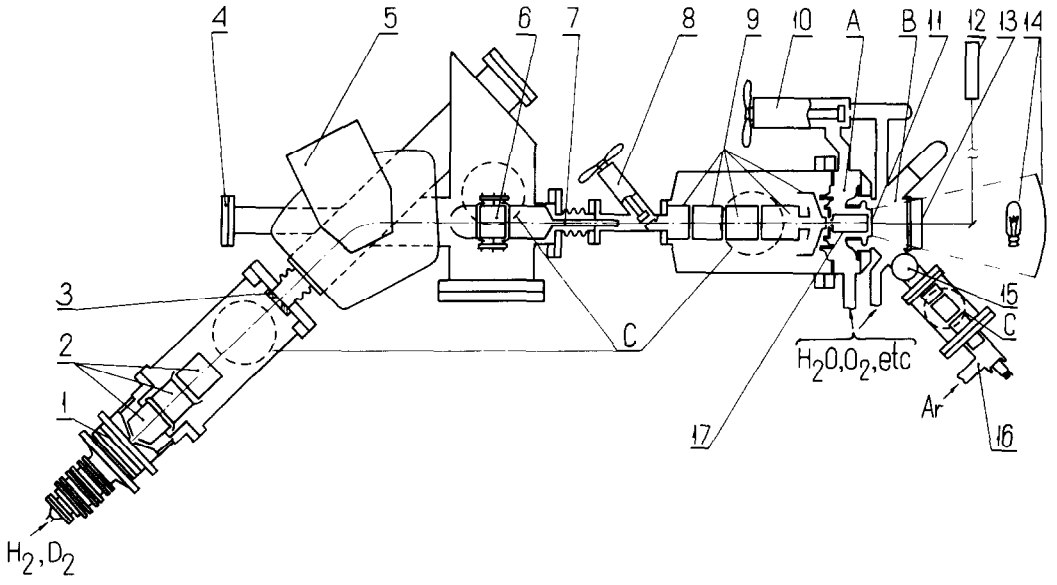



Fig. 2. Lay-out of the experimental apparatus: (1) duoplasmatron, (2) 1st electrostatic lens, (3) diaphragm, (4) flange on the line of sight with the membrane (where an atomic beam source can be adjusted), (5) sector magnet, (6) 2nd electrostatic lens, (7) tube diaphragm, (8) valve, (9) electrodes of the 3rd electrostatic lens and of the electrostatic retardation system, (10) connection valve between the inlet and outlet chambers, (11) membrane sample, (12) laser (to check the radial distribution of hydrogen in the sample by local heating), (13) plane glass to let in the light heating the membrane, (14) lamp and reflector to heat the membrane, (15) magnet-driven shutter, (16) source of $Ar^+$ ion beam, (17) cylinder "collector"; (A) inlet chamber, (B) outlet chamber, (C) pumping ducts to the ion pumps (cross-section).

operating with low energy ions) [2]. The ion current of $H_2^+$- or $H_3^+$-ions amounts to about 100 $\mu$A. With focusing the ion beam into the round 7–12 mm diameter spot on the sample plane, we get the flux density of the order of $10^{15}$ proton $cm^{-2}$ $s^{-1}$, the non-uniformity of density over the spot being within the 5–10% limit. The density of 20 keV $H_3^+$ ion beam can be risen to about $10^{17}$ proton $cm^{-2}$ $s^{-1}$ by focusing it into a smaller spot.

### 2.3. Vacuum

Four 250 l/s ion pumps serve to pump out the apparatus. The upstream and the downstream chambers are baked by external heating so that the residual pressure in the absence of the ion beam does not exceed $10^{-7}$ Pa (mainly CO, then $H_2O$).

### 2.4. Sample treatment in situ

The upstream surface of a membrane sample can be cleaned due to the bombardment by the working ion beam from duoplasmatron, while a special $Ar^+$-ion source is set to clean the downstream surface (5–10 $\mu$A, 500–800 eV). In addition, refractory metal membranes can be cleaned by electron bombardment heating to whatever high temperature permissible for any particular material. For this purpose, a special movable unit is provided (5 in fig. 3) consisting of a thick 2 cm diameter disk cathode made of the same material as the membrane sample (to avoid the evaporation of foreign species) and heated from behind by the electron bombardment from a tungsten filament cathode. To avoid the evaporation of material onto the plane optical glass during the membrane heating procedures, a protecting shield driven by an external magnet may be slid between the glass and the membrane.

In the course of the ion-beam experiment the membrane is heated by the light from external source (a spot-light filament lamp) focused through the plane glass on the membrane by a reflector. The image of the incandescent filament is spread over the whole membrane area (4 cm diameter), so that the temperature in the central part where the ion beam hits the membrane (about 1 cm diameter) proves to be virtually uniform.

The temperature of the sample is not measured

[2] The operational modes relating to the ion retardation will be discussed in more detail elsewhere. Further in the present paper, we confine ourselves to the experiments with 20 keV ions only.

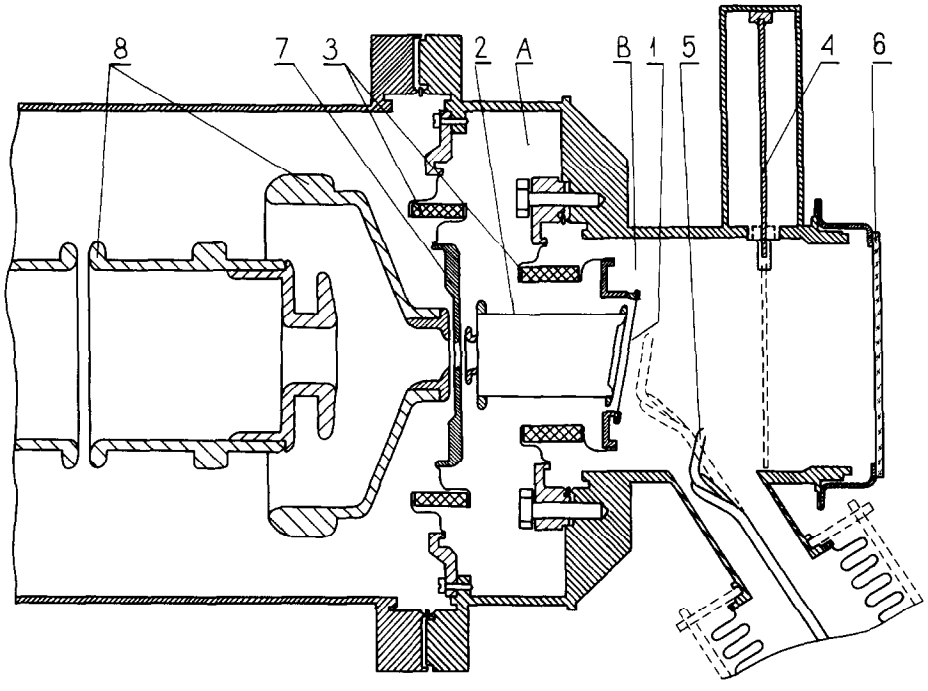



Fig. 3. Sketch of the part of apparatus adjacent to the membrane sample. (1) membrane, (2) cylinder "collector" (feed-throughs and supports are not shown), (3) ceramic insulators, (4) magnet-driven protecting shield, (5) cathode unit driven by bellows for the high-temperature heating of the membrane, (6) plane optical glass, (7) a partition with the inlet diaphragm separating the inlet chamber, (8) retardation system electrodes; (A) inlet chamber, (B) outlet chamber.

directly during membrane experiments, and a model sample, with several thermocouples spot-welded to it, was to be substituted for the working membrane [3] in order to scale temperature for all modes of operation. The additional heating of the central part of the membrane due to 100 $\mu$A ion beam of 20 keV was measured to be 80–120 K (depending on an initial temperature); about 90% of this addition is established in 10–15 s upon switching the beam.

Controlled coverage of either surface of the membrane by non-metallic films [8,18,19] can be effectuated by reactions with chemically active gases, such as $O_2$, $H_2O$, $C_2H_2$, $C_6H_6$, $H_2S$, etc. These are admitted into either upstream or downstream chamber through leak-valves, the partial pressures being measured by the omegatrons. Besides, thermal evaporation can be employed for covering the downstream surface with metal overlayers, e.g. with palladium.

### 2.5. Collector of secondary hydrogen particles

Placed adjacent to the membrane sample and co-axial with the ion beam is a cylinder (17 in fig. 3) of 10 $\mu$m Nb foil, 3 cm diameter, 5 cm long, and having a heading on the entrance side with a 1 cm diameter orifice to let in the beam. Its purpose is to catch and collect secondary supra-thermal hydrogen particles: both those reflected by the membrane, and re-emitted under the action of ion beam. The resistance heating of the collector by dc is available up to about 1300 K to release the accumulated hydrogen. The membrane sample is adjusted slanting against the normal to the ion beam axis (as seen in fig. 3) in order that the maximum of the reflected particle angular distribution gets to the collector walls rather than to the aperture of the inlet diaphragm.

The reflection of ions and the ion-induced re-emission are not important for the experiments with 20 keV $H_2^+$-ions described in the present paper. It is hoped, however, that we shall be able to estimate the reflection coefficient with the use of collector in experiments with low energy ions planned for the future, and to get some information about the energetic distribution of hydrogen particles released through ion-induced re-emission [41].

[3] The temperatures above 900 K can also be measured by a pyrometer directly on the working sample, using a window mounted on the flange on line of sight with the membrane [(4) in fig. 2].

The reflection of hydrogen particles exceeds 50% for the case of not too high energies (tens to hundreds of eV) [42]. Such energies are expected, for instance, in the divertor plasma of the ITER tokamak. Consequently, the trapping probability at one impact may not be sufficient to provide for the effective pumping of these particles. The combination of the membrane target with the cylinder collector in this setup delivers a model of the pump where several collisions with absorbing walls may take place, and the efficacy of such a multi-stroke pump can be directly estimated. Notice that pumping by collector can be "switched off" by rising its temperature to a value (> 900 K) with which hydrogen inventory in the niobium foil becomes negligible.

### 2.6. Measuring the absolute rate of sorptional and membrane pumping

The problem here lies in precisely determining the value of the pressure originating from hydrogen transported by the ion beam into the inlet chamber when both the membrane and the sorptional pumping are absent. This becomes particularly important with the ion beam retardation to lower energies where the ion current measurements are less reliable because of the re-distribution of secondary electron currents between the target and the retarding electrodes.

The solution is found in measuring the hydrogen pressure in the upstream chamber when, in the regime of steady-state re-emission and permeation, the whole flow of gas penetrating the membrane is returned back to the inlet side through the open valve between the up- and the downstream chambers (see fig. 2), while the pump of the downstream chamber is cut off by a magnet-driven shutter [(15) in fig. 2].

In addition, an operational mode is easily reached (as will be seen from the experimental data below), when the steady-state permeation flow is much smaller than the steady-state re-emission. Thus practically zero pumping by the membrane is established, with the whole flow of implanted hydrogen being returned to the upstream chamber even without opening the connection valve and shutting off the downstream pump.

## 3. Experimental data and discussion

### 3.1. The mode of the experiment

A niobium membrane, 0.01 cm thick and 4 cm diameter, is fixed by argon-arc welding. It was degassed in ultra-high vacuum through heating by a light to 950

K for about 25 h. The downstream surface was not subject to any further treatment, and, undoubtedly, it is covered by non-metal contaminants. AES experiments performed on a niobium specimen of the same batch have shown that the main foreign species present on the surface and in the near-surface region are oxygen and carbon. Heating the specimen to 1000 K in ultra-high vacuum results in a decrease (4 to 5 times) in the contaminant concentration, apparently, due to their dissolution in the metal lattice. The resulting concentrations of oxygen and carbon coincide in the order of magnitude.

The "natural" downstream surface in our experiments is, therefore, believed to be passivated with oxygen/carbon coverage, possibly, with the mixture of surface oxides and carbides. The probability of dissolution for a thermal hydrogen molecules, $\alpha$, strongly depends on metal temperature in the case of "natural" passivated surface [8,25,27], and it is orders of magnitude less than for clean niobium surface, the latter being characterized by $\alpha$ of the order of about 0.1 with a very weak temperature dependence [6,43].

The beam of 20 keV $H_2^+$-ions was employed in these experiments, which is equivalent to a 10 keV $H^+$-beam of double intensity. The beam was focused into a $\sim 0.8$ cm diameter spot on the membrane, with the ion current of 60 $\mu$A, i.e. $1.5 \times 10^{15}$ proton $cm^{-2}$ $s^{-1}$. The secondary electron current was suppressed by biassing the collector and the inlet diaphragm [(2) and (7) in fig. 3] $-250$ and $-500$ V respectively relative to the membrane target [4]. It had been demonstrated, with the use of fluorescent quartz target and a micro-Faraday-cup probe, that the current density was virtually uniform over the whole spot, with practically no ion current outside it.

The experimental procedure was as follows. First the membrane temperature and the pressure of admitted gas ($H_2O$ or $O_2$) were fixed [5]. The operational flux of hydrogen was admitted into the ion source, and a background hydrogen pressure was established in the upstream chamber caused by the influx of hydrogen from the duoplasmatron. After that, the $H_2^+$ ion beam was deflected by the magnet into the diaphragm of upstream chamber (see figs. 2 and 3), and the processes of hydrogen re-emission and permeation were being monitored by registering hydrogen pressure in the chambers on the both sides of the membrane. On reaching the steady-state conditions, the admission of $H_2O$ or $O_2$ was ceased, the ion beam was switched off, and the membrane temperature was raised to 970 K in order to accomplish degassing the membrane.

The whole procedure was then repeated with a higher $H_2O$ or $O_2$ pressure. After the experiments with higher pressures, the results for lower pressures could be reproduced only following a prolonged ion bombardment of the upstream surface in ultra-high vacuum.

### *3.2. Absorption, re-emission and permeation in the presence of $H_2O$ and $O_2$*

Fig. 4 represents the build-up with time of the re-emission flux, $j_0$, and of the permeation flux, $j_L$, after switching on the ion beam at different pressures of $H_2O$ and $O_2$ present on the upstream side, and at different membrane temperatures. These plots demonstrate the pumping of $H_2^+$-ions by the membrane: initially the sorptional pumping occurs, and later, on reaching the balance between the implanted ion flow and the flows of hydrogen released to the both sides of the membrane, the stationary membrane pumping is established.

At the lowest $H_2O$ and $O_2$ pressures, the re-emission flux, $j_0$, rapidly reaches its steady-state magnitude, $j_{0\infty}$, which is virtually equal to the incident flux of ions, $j_i$, for all the temperatures represented (838, 623 and 500 K). The steady-state permeation flux is small: $j_{L\infty} \ll j_i$. The increase of $H_2O$ or $O_2$ pressure first leads to drastic deceleration in reaching the steady-state re-emission, and to rise in permeability, though the relations $j_{0\infty}/j_i \approx 1$, and $j_{L\infty}/j_i \ll 1$ remain true [6]. Thus the efficiency of sorptional pumping becomes considerably higher, while no appreciable membrane pumping is developed. With the further increase of $H_2O$ or $O_2$ pressure, the steady-state permeation rises further, so that at first an appreciable part, and then the major part of the inci-

[4] Measurements with other bias configurations show that the current of secondary electrons from collector does not add more than 5% to the current measured in the membrane target circuit.

[5] Two remarks are necessary here. First, the spot where the ion beam hits the membrane gets an additional heating by about 100 K (see section 2.4) established in tens of seconds after switching the beam; it is this established temperature indicated by the experimental data below. Second, the exposition of the membrane sample at the operational pressures of $H_2O$ or $O_2$ for the period of time exceeding the duration of any single experimental run by an order of magnitude did not change the following results.

[6] For $j_{L\infty}/j_i \ll 1$, the changes in permeability are scarcely seen in fig. 4 because of the linear scale used. In reality, the changes caused by admitting $H_2O$ or $O_2$ may be in orders of magnitude, which is easily registered by the omegatron (it is possible to measure $j_L \approx 10^{-4} j_i$).

dent ion flux penetrates the membrane. In other words, first an appreciable stationary pumping by the membrane takes place, and then superpermeability develops (e.g. $j_{0\infty}/j_i \approx 0.25$ and $j_{L\infty}/j_i \approx 0.75$ is observed at 838 K with a $H_2O$ pressure of $1.3 \times 10^{-3}$ Pa).

When one stops admitting $H_2O$ or $O_2$, it brings about rather a spectacular effects (see fig. 5): with the upstream surface sputtering by the hydrogen ion beam being continued, the re-emission flux instantly begins to rise while the permeation flux drops. The re-emission flux passes through a maximum with $j_0/j_i > 1$ owing to the evolution of the hydrogen taken up by the membrane during the period when the evolution through the upstream surface has been blocked. Interestingly, the re-emission build-up takes rather a large period of time after the admittance of $H_2O$ or $O_2$ having been ceased:

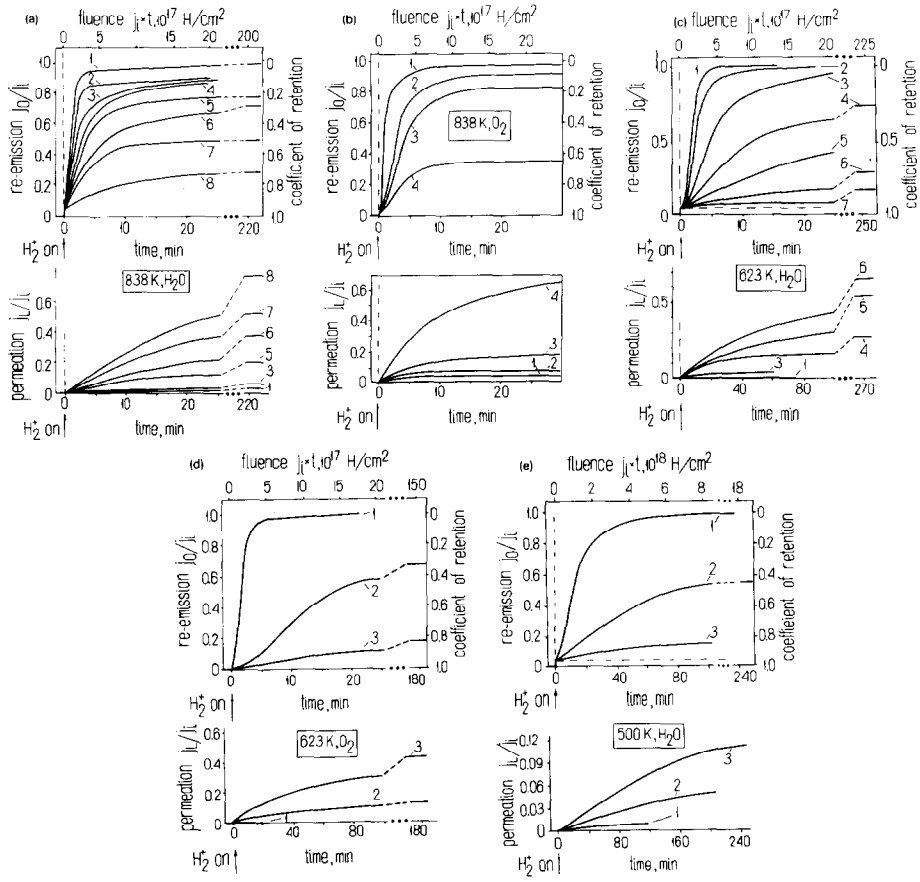



Fig. 4. Effects of $H_2O$ and $O_2$ present in the gas phase on the upstream side of the membrane upon the transient and the steady-state re-emission and permeation of hydrogen. The niobium membrane, 0.01 cm thick, is acted upon by 20 keV $H_2^+$ ion beam with the ion flux of $7.5\times10^{14}$ $H_2^+$/(cm$^2$ s) (which corresponds to the 10 keV $H^+$-flux of $1.5\times10^{15}$ $H^+$/(cm$^2$ s)). The outlet surface of the membrane is "natural", i.e. it is passivated by residual non-metallic contaminants. Varying from one curve to another in each plot is the pressure of $H_2O$ or $O_2$, with the ratio of the flux of O atoms (in $H_2O$ or $O_2$ molecules) onto the surface to the flux of protons (in $H_2^+$ ions) indicated in the brackets. Note that the metal temperature rises by about 100 K in first 10–15 s after switching the ion beam to reach the steady state value indicated by the plots. (a) Membrane temperature 838 K. $H_2O$ pressure: (1) $2\times10^{-7}$ Pa (0.0005), (2) $1.3\times10^{-6}$ (0.003), (3) $5\times10^{-5}$ (0.01), (4) $1.2\times10^{-5}$ (0.03), (5) $4\times10^{-5}$ (0.1), (6) $1.3\times10^{-4}$ (0.3), (7) $4\times10^{-4}$ (1), (8) $1.3\times10^{-3}$ (3). (b) Membrane temperature 838 K. $O_2$ pressure: (1) 0, (2) $1.8\times10^{-6}$ Pa (0.007), (3) $5.5\times10^{-6}$ (0.02), (4) $6.5\times10^{-5}$ (0.25). (c) Membrane temperature 623 K. $H_2O$ pressure: (1) $3.2\times10^{-7}$ Pa (0.0008), (2) $1.6\times10^{-6}$ (0.004), (3) $5.7\times10^{-6}$ (0.015), (4) $1.3\times10^{-5}$ (0.03), (5) $4\times10^{-5}$ (0.1), (6) $1.6\times10^{-4}$ (0.4), (7) $4.8\times10^{-4}$ (1.2). (d) Membrane temperature 623 K. $O_2$ pressure: (1) 0, (2) $7.7\times10^{-6}$ Pa (0.03), (3) $2.2\times10^{-5}$ (0.085). (e) Membrane temperature 500 K. $H_2O$ pressure: (1) $1\times10^{-6}$ Pa (0.003), (2) $3.2\times10^{-6}$ (0.008), (3) $1.3\times10^{-5}$ (0.03).

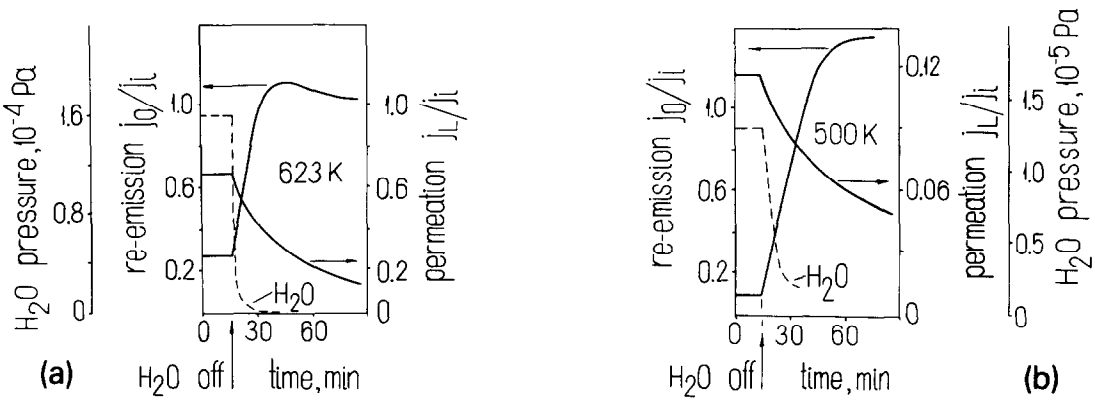



Fig. 5. The changes of re-emission and permeation fluxes resulting from the cessation of $H_2O$ being admitted onto the inlet side after the establishment of steady-state re-emission and permeation (with the $H_2^+$-ion beam unchanged). The equivalent flux of protons is $1.5\times10^{15}$ $H^+/(cm^2\ s)$, with the energy of 10 keV; the niobium membrane thickness is 0.01 cm. The change of $H_2O$ pressure with time is shown by the dashed curves. (a) Membrane temperature 623 K, initial $H_2O$ pressure $1.6\times10^{-4}$ Pa. (b) Membrane temperature 500 K, initial $H_2O$ pressure $1.3\times10^{-5}$ Pa.

scores of minutes are required, whereas the time of sputtering one monolayer oxygen is in the order of one minute. It may be concluded that the oxidation in the process of ion bombardment does not only affect the surface, but also subsurface layers.

### 3.3. Superpermeability

Two requirements are necessary for superpermeability to take place [1,2,8]: (i) the upstream surface should effectively return back to bulk metal the implanted hydrogen particles approaching it from the inside by way of diffusion, so that virtually every hydrogen atom absorbed reaches the downstream boundary before being re-emitted (the "strong reflection" requirement, in the terminology of refs. [1,2,8]); (ii) there is a limitation on the symmetry of desorption requiring that a hydrogen atom reaching the downstream boundary should be released through it with a probability not less than that of re-emission when approaching the upstream boundary.

It is easy to show that lest the re-emission is determined by ion-induced processes, the first requirement is readily met in the H–Nb (V, Ta) system. This requirement is satisfied here even for the clean surfaces, with zero $\mathscr{E}_0$ barrier. The dissolved hydrogen atoms are returned back from the solid-gas interface into the metal lattice owing to that the energy level $\mathscr{E}_p$ (see fig. 1) for the H–Nb (V, Ta) system is much lower (by about 30 kJ/mol H) than the level of H atom in a free $H_2$ molecule, unlike, e.g., in the case of H–Pd system.

Anyway, the H atoms absorbed did not only reach the downstream surface, but repeatedly crossed the membrane by diffusion in the experiments described below where the upstream surface was being continuously sputter-cleaned by a 20 keV $H_2^+$-ion beam. The fact was evidenced by the characteristic time for the re-emission build-up, $\tau_r$, having been much larger than the diffusion time $L^2/6D$ ($D = 5\times10^{-4}\exp(-10450$ J/mol H$/RT)$ $cm^2/s$ for H in Nb [5]).

As for the second requirement concerning the symmetry of evolution, the possibility to satisfy it, with the upstream surface being continuously ion-sputtered and the downstream one retaining the "natural" passivation, is not so evident a priori. Nevertheless, superpermeability has been reached in our experiments at the temperatures of 623 and 838 K (fig. 4). This appears to testify of an insignificant role played by ion-induced evolution, at least for niobium surface passivated by oxygen at $T \geqslant$ 623 K. One may also conclude that the passivation by oxygen appears to be more effective than the "natural" passivation of downstream surface in suppressing the thermal evolution of hydrogen.

### 3.4. Dynamics of surface passivation

One can see from the above that a considerable deceleration of the re-emission build-up manifests itself at the densities of O-atom flux (in $H_2O$ or $O_2$ molecules) falling on the surface that are nearly 100 less than the proton flux from the $H_2^+$-ion beam. It would seem likely to expect that a significant effect in the suppression of re-emission may be gained with an oxygen coverage of no less than about one monolayer. This

implies that the sticking coefficient of both $H_2O$ and $O_2$ for the process of niobium surface oxidation is approximately 100 times larger than the proton sputtering yield. We are unaware of the precise data concerning the coefficients of oxygen sputtering from $Nb_mO_n$ oxides by hydrogen ions, though, judging by the data reviewed by Betz and Wehner [44], the sputtering yield for 10 keV protons would be in the order of $10^{-2}$ (chemical sputtering of the oxygen strongly bound in niobium oxides would seem highly improbable, very much like in the case of aluminium [45]).

Consequently, the sticking coefficient of $O_2$ and $H_2O$ molecules on the Nb surface during the process of surface sputtering by protons appears to have the value of the order of unity. This corresponds to the known data referring to the kinetics of $O_2$ chemisorption on the clean niobium surface [46], and on the surface being sputtered by $Ar^+$-ions [47]. At the same time, re-emission is more effectively suppressed by the action of $O_2$ than by $H_2O$ corresponding to the same flux of O-atoms on the surface at 838 K, as one can see from comparing figs. 4a and 4b. This suggests that the sticking coefficient of $O_2$ may be greater [7] than that of $H_2O$, at least at the temperature of 838 K.

The sputtering by protons in our experiments should, evidently, create and maintain a considerably high concentration of "holes" in the oxide coating. Nevertheless, this coating is very effective in the prevention of the back evolution of implanted hydrogen, giving thus rise to superpermeability on a membrane with a "natural" passivated surface downstream. This fact can be explained by the alternating valency of niobium in the oxides. The surface characterized by a relatively valency-saturated state (e.g. $Nb_2O_5$) may not become metallized after discharging oxygen, but may change into less oxidized states that retain, at least partly, the passivating properties with respect to hydrogen evolution (i.e. $\mathscr{E}_0$ barrier remains effectively non-zero, see fig. 1). An increase of $O_2$ or $H_2O$ flux, and the corresponding formation of more saturated oxides, may, on the other hand, result not only in a lower concentration of "holes" in the coating, but also in a higher $\mathscr{E}_0$ barrier. The transformation of $Nb_mO_n$ oxides into different forms with varying the $O_2$ pressure and the rate of sputtering by $Ar^+$-ions was observed by Koval et al. [47].

[7] Special precautions have been taken in these experiments to correctly measure the oxygen pressure by the omegatron having a hot tungsten cathode.

### 3.5. Temperature dependence of the effects of $H_2O$

Fig. 4 shows that with equal $H_2O$ pressures, the lower the membrane temperature, the larger is the time $\tau_r$ for the re-emission build-up, the greater the steady-state sorption capacity $C_\infty$, and the larger the permeation probability $j_{L\infty}/j_i$. As regards $\tau_r$ and $C_\infty$, this can be explained as follows. The coating by oxygen, on which the barrier $\mathscr{E}_0$ depends, seems to be little affected by temperature, as the sticking coefficient of $H_2O$ is temperature independent. On the other hand, the lower the temperature, the more effectively the evolution of hydrogen is hindered by the surface barriers. Now, if $\mathscr{E}_0 > \mathscr{E}_L$ takes place (i.e. if the barrier created by oxygen on the surface continuously bombarded by protons is higher than that characteristic of the "natural" downstream surface), then the influence of $H_2O$ on permeability will be the stronger, the lower the temperature.

Interestingly, the opposite effect was observed in similar experiments on austenitic stainless steel bombarded by $D^+$ ions in the presence of $O_2$ and $H_2S$ on the upstream side of the membrane: a rise of temperature enhanced the influence of $O_2$ and $H_2S$ on hydrogen uptake and permeation [8,17]. This may be connected with a significant increase with temperature of the sticking coefficients of $O_2$ and $H_2S$ on the stainless steel surface.

The possibility of oxygen dissolving in the bulk metal with the rise of temperature should also be taken into consideration (see section 3.2), but there is some evidence to confirm that this process may become significant only at temperatures higher than 1300 K for niobium [46,47].

### 3.6. Possible effects of radial diffusion

Two points should be singled out in the data presented above requiring special elucidation:

(A) It takes a longer period of time for the permeation flux to reach the steady-state magnitude than for the re-emission flux, whereas these periods should coincide for the regime of "strong reflection" (see section 3.3), as it follows from both the theoretical treatment [1–3,48], and the experimental data [12].

(B) The balance of fluxes ($j_{0\infty}/j_i + j_{L\infty}/j_i = 1$) is not actually reached with the apparently stationary regimes for both the re-emission and permeation fluxes at the temperature of 500 K in the presence of oxygen (fig. 4e)[8].

[8] The disbalance, though much less conspicuous one, is observed at 623 K as well (figs. 4c and 4d).

According to our estimates, both the points (A) and (B) are accounted for by the radial diffusion of implanted hydrogen from the spot where the ion beam hits the membrane (the spot radius $r_b$ is 0.4 cm) to the membrane periphery (the membrane radius is 2 cm). It should be noted, however, that it is the area of $r \leqslant r_b$ where the evolution of hydrogen mainly occurs, since this area gets an additional heating by the beam (which is mostly essential at lower temperatures), and it is cleaned by the beam (which is essential at lower $H_2O$ or $O_2$ pressures). That the diffusion into the region of $r > r_b$ may play a crucial role in the observed transitional re-emission has been shown by Hotston and McCracken [33] in their experiment on a niobium target of the thickness $L = 0.1$ cm, with the ion beam radius of $r_b = 0.15$ cm. The corresponding figures are more beneficial in our experiments ($L = 0.01$ cm and $r_b = 0.4$ cm) so that the radial diffusion effects should be less pronounced. Still our estimates (in agreement with those by Hotston and McCracken) confirm that the radial diffusion may have considerably affected the transitional re-emission and permeation under the conditions of our experiments. These effects are due to a remarkably high mobility of dissolved hydrogen in Nb (V, Ta) coupled with the large negative heat of solution.

One might also consider hydrogen trapping in radiation defects to be responsible for the disbalance of fluxes (i.e. for the point (B) above). The research by Hotston and McCracken, however, may be referred to here again. These authors had no need to consider the radiation damage effects in order to get a very good agreement between their calculations and experimental data.

### 3.7. Re-emission at 400 K

At this temperature, the $\beta$-phase of hydrogen solution in niobium may be formed [5]. Therefore, only UHV conditions were employed for the ion-beam experiment, lest the surface passivation by oxygen would lead to a too high hydrogen concentration, and to irreversible changes of the sample structure. The development of re-emission with time at 400 K (fig. 6) exhibits the same trend to the uptake of hydrogen being enhanced with the temperature decrease. This agrees well with the data found in the literature [49] where the maximum uptake by niobium was observed around 300 K. Nevertheless, a number of questions is set by the data presented here.

Hotston and McCracken found it possible to treat the uptake and re-emission of hydrogen in the terms of their model that included only the thermal evolution

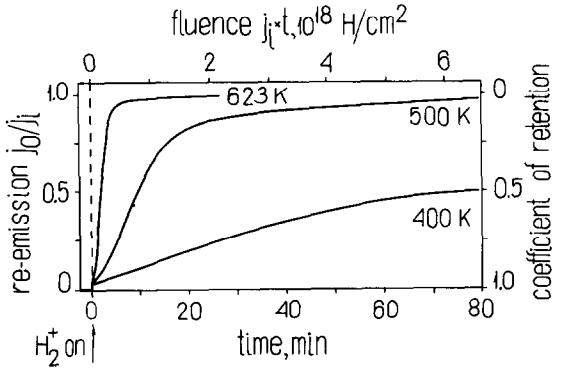



Fig. 6. Transient re-emission with ultra-high vacuum on the inlet side (no gases specially admitted). The equivalent flux of protons is $1.5 \times 10^{15}$ $H^+/(cm^2\ s)$, with the energy of 10 keV; the niobium membrane thickness is 0.01 cm.

according to the equation $j_0 = k_{r0}C^2$, and the thermal diffusion, even at the temperature as low as 435 K [33]. But the re-emission in their experiments did, in fact, never start during the experimental run, so they had to explain only the absence of re-emission with a particular dose of implanted hydrogen. In our experiments, we are also able to interpret the results at $T \geqslant 500$ K referring solely to the thermal re-emission process. But we cannot justify the re-emission flux of $7 \times 10^{14}$ atom H/(cm$^2$ s) at 400 K (as in fig. 6) in the same way (the fluxes are counted in atoms H throughout the paper for the sake of uniformity of notation, though the evolution flow consists, of course, of $H_2$ molecules). Indeed, assuming the surface to be clean under the action of ion bombardment in ultrahigh vacuum at 400 K, we can express the maximum flux that may be evolved through the surface covered by a monolayer of chemisorbed hydrogen as

$$j_{0\max} = 10^{15} \times \exp(-2\,\Delta H_s/RT) \quad \text{H atom/(cm}^2\ \text{s)},$$

here $\Delta H_s$ is the enthalpy of chemisorption at the sites via which the evolution of dissolved hydrogen goes. According to Pick [6,43], $\Delta H_s = -56.4$ kJ/mol H for a monolayer coverage, while our own data (unpublished) give $\Delta H_s = -75$ kJ/mol H [9]. Taking $\Delta H_s = -56.4$ kJ/mol H for the maximum estimate of the desorption flux, we get $j_{0\max} \approx 3 \times 10^{13}$ H atoms/(cm$^2$ s), a value 10 times less than that observed experimentally. An

[9] The data were obtained in experiments with a 10 $\mu$m thick Nb ribbon cleaned by heating to 2400 K. The value of $\Delta H_s$ was deduced from the data on the rate of hydrogen dissolution through the surface saturated by chemisorbed hydrogen (very much like in Pick's experiment [43]).

attempt could be made to attribute the observed re-emission to ion-induced desorption, whose coefficient should amount to $\sim 0.5$ $H_2$ molecule per $H_2^+$-ion in such a case. This large value may seem rather dubious for the sputtering of chemisorbed atoms by 10 keV protons. As for the contribution of near-surface hydrogen atoms into the ion-induced evolution, it should not be appreciable, if one assumes the implanted H atoms to be distributed uniformly in the bulk of the metal owing to their high mobility (viz., their concentration in the lattice would not exceed 0.01 H/Nb).

The plausible explanation may be that the distribution of implanted hydrogen is not homogeneous in the bulk metal. The radiation defects are not, apparently, annealed at 400 K [50], so the concentration of defects for the 10 keV proton fluence of $10^{18}$–$10^{19}$ $H^+/cm^2$ may become very greate in the zone of implantation. These defects may serve as effective traps for hydrogen atoms at 400 K, so one is inclined to suggest that the re-emission observed should be ascribed to the ion-induced desorption from the sub-surface region enriched by hydrogen because of the high concentration of radiation defects.

### 3.8. The rate constant for thermal evolution, $k_{r0}$

The $k_{r0}$ constant characterizing the surface continuously bombarded by $H_2^+$ ions may be evaluated from our data at different metal temperatures and at different pressures of $H_2O$ and $O_2$. The evaluation proceeds from the following assumptions: (i) re-emission is a thermally controlled process at $T \geqslant 500$ K (see section 3.7); (ii) re-emission obeys the law $j_0 = k_{r0}C^2$; (iii) the concentration of dissolved hydrogen, $C$, is virtually uniform throughout the bulk at any moment (the time $L^2/6D$ required for equalizing the concentration is less than 0.5 s at $T \geqslant 500$ K for $L = 0.01$ cm).

There are different ways to evaluate $k_{r0}$. One way is to use the relation

$$k_{r0} = j_0(t)/C^2(t) = j_0(t)L^2/(j_i t)^2,$$

valid for the time moments $t \ll \tau_r$ ($\tau_r$ is the characteristic time of the re-emission build-up) when virtually all the hydrogen implanted is still retained in the membrane. All the quantities in this equation are directly provided by the data in fig. 4 (The flux $j_0(t) \ll j_{0\infty}$ is easily measured on more sensitive scales.) Some difficulty arises here in that some time, $\tau_s$, is required for the surface composition to reach the steady state under the combined action of the active gas and of the sputtering ion beam. The time, $t$, is to be taken within the range $\tau_s \ll t \ll \tau_r$, and the method is only applicable in the case of $\tau_r \gg \tau_s$, e.g. at 500 K.

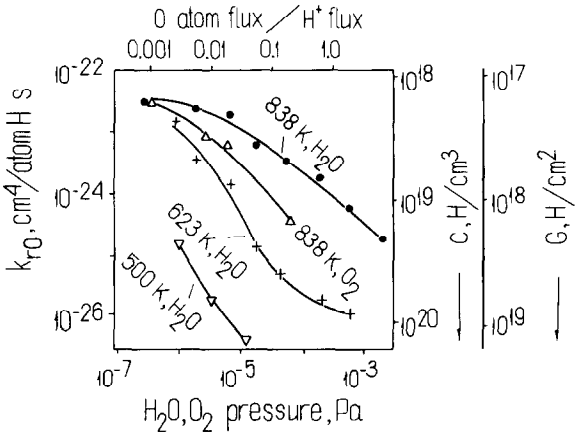



Fig. 7. The rate constant, $k_{r0}$, for the thermal re-emission of hydrogen by niobium continuously acted upon by $H_2^+$ ion beam, as a function of the pressures of $H_2O$ and $O_2$. The equivalent flux of protons is $1.5 \times 10^{15}$ $H^+/(cm^2\ s)$, with the energy of 10 keV ($k_r$ may depend on ion energy via the rate of sputtering the passivating films). The axes on the right are scaling the concentration of hydrogen, $C$, that niobium accumulates before the beginning of appreciable (with $\Psi = 0.1$) re-emission, and the corresponding dose, $G$, that a "thin" 0.1 cm Nb target having an impermeable outlet surface is able to absorb.

Another way is to first determine $k_{rL}$, the rate constant of evolution through the downstream surface, by the equation

$$k_{rL} = j_L(t)L^2/(j_i t)^2$$

valid at $t \ll \tau_r$. Unlike $k_{r0}$, the rate constant $k_{rL}$ is not changing after the ion beam initiation, and it is not influenced by the presence of $H_2O$ or $O_2$ on the upstream side. Having found $k_{rL}$, one obtains $k_{r0}$ for the steady-state permeation regime from

$$k_{r0} = k_{rL} j_{0\infty}/j_{L\infty},$$

where $j_\infty$ and $j_{L\infty}$ are the steady-state fluxes of re-emission and permeation respectively (see fig. 4). This method is not perfect either because of the radial diffusion effects; for instance, it is not suitable for the data obtained at 500 K (see section 3.6).

Both these methods were used in this work to get the $k_{r0}$ values, the choice between them depending on specific circumstances. The results obtained are summarized in fig. 7.

These data may be compared to those predicted for the clean surface. Note that $k_r$ is related to $\alpha$, the sticking coefficient for thermal $H_2$ molecules to get into the state of dissociative binding ($\alpha = \alpha^* \exp(-2\mathscr{E}_0/$

$RT$), fig. 1), via the Sieverts' constant, $S$ ($S = S^* \exp(-\Delta H_v/RT)$, fig. 1), by the equation [10]

$$k_r = \frac{2Z\alpha}{S^2} = \frac{2Z\alpha^*}{(S^*)^2} \exp(-2\epsilon_0 + 2\Delta H_v)/RT, \qquad (1)$$

where $Z \approx 1.1 \times 10^{19}$ molecule $H_2/(cm^2\ s\ Pa)$ is the kinetic theory coefficient. For niobium, $S = 7 \times 10^6 \exp(39700\ J/mol\ H/RT)$ atom $H/(cm^3\ Pa^{1/2})$ [5], while $\alpha^* \approx 0.1$, and $\mathscr{E}_0 \approx 0$ are expected for the clean surface [6,9]. Thus we get by this evaluation $k_{r0} = 3 \times 10^{-24}$, $1 \times 10^{-22}$ and $6 \times 10^{-21}\ cm^4\ (atom\ H)^{-1}\ s^{-1}$ for clean niobium surface at 500, 623 and 838 K respectively.

One can see from fig. 7 that the values of $k_{r0}$ obtained experimentally at the lowest pressures of $H_2O$ and $O_2$ are somewhat smaller than those predicted for the clean surface, but the trend is seen for $k_{r0}$ to be continuing to rise with the $H_2O$ pressure decrease even in the pressure range below $10^{-6}$ Pa at the temperatures of 500 and 623 K. There is a more marked disagreement at 838 K, and in particular so, as the rise of $k_{r0}$ with the decrease of $H_2O$ pressure ceases over the pressure range of $10^{-6}$–$10^{-7}$ Pa at this temperature. This mismatch bears a very likely explanation however that the segregation of non-metal contaminants out of the niobium lattice to the surface becomes significant at $T \geqslant 800$ K, so the surface, even being continuously sputtered in ultra-high vacuum, may be covered by contaminant films permanently restored [11].

We would like to warn against a simplistic approach to interpretations in the terms of eq. (1) of the temperature dependence of $k_{r0}$, as produced by these experiments. The point is that the values of both the pre-exponent $\alpha^*$ and the activation barrier $\mathscr{E}_0$ may depend on metal temperature, because the state of the surface is determined by a dynamic balance between such processes as ion sputtering, oxidation through the interaction with $H_2O$ or $O_2$ molecules from the gas phase, oxygen dissolution in bulk metal and its back precipitation toward the surface, and also the mutual transformation of different forms of niobium oxides (see sections 3.4 and 3.5) on the surface and in the nearest sub-surface region. At least the two latter processes are strongly dependent on temperature. More scrupulous experiments would be needed to elucidate this complex physico-chemical situation.

[10] The formula [2,5,8,21] is obtained by combining Sieverts' law, $C_e = S\sqrt{p}$, and the balance equation, $2Zp\alpha = k_r C_e^2$, where $C_e$ is the equilibrium concentration of hydrogen dissolved in metal corresponding to the hydrogen pressure, $p$ (the factor 2 appears because the flux of $H_2$ molecules is calculated in the number of atoms). Note that eq. (1) is valid for the metals occluding hydrogen both exothermically and endothermically.

[11] Special experiments were carried out to follow the decrease of the probability of $H_2$ molecules dissociative binding, $\alpha$, due to the segregation of contaminants onto the surface of a niobium sample (taken from the same batch as in these experiments) that had been previously cleaned by ion sputtering. The flux of contaminants onto the surface was found to be $\sim 2 \times 10^{13}$ atom/($cm^2$ s) at 810 K, whereas at 670 K it did amount only to $\sim 1 \times 10^{12}$ atom/($cm^2$ s).

## 4. The pumping of fast hydrogen particles

The purpose of this section is two-fold. For one thing, we wish to formulate a general approach to the estimates of sorptional and membrane pumping based on experimental data like those presented above. Second, we are going to bring out some particular estimates pertaining to the conditions of the present experiments in order to demonstrate the possibilities that the dynamic passivation of surface render for pumping, for compression, and for the separation of hydrogen from helium.

### *4.1. Sorptional pumping*

Drawing estimates of the sorptional pumping capacity directly from the data of the present membrane experiments may be incorrect. Indeed, one sees from fig. 4 that a considerable, and, at higher pressures of the passivating gases, even the major part of the implanted hydrogen is being evolved not by re-emission, but via the outlet surface of the membrane target. The big share of the downstream evolution is not, it should be emphasized, a consequence of a comparatively thin sample having been used. The distribution of the evolution fluxes between the two surfaces, $j_0(t)/j_L(t)$, is virtually independent of the membrane thickness up to the thicknesses measuring in ones and tens of centimetres, as will be shown below, and till up these thicknesses the share of the downstream evolution will be the same as in our experiments (we remind that the larger time required for the permeation flux than for the re-emission flux to reach the steady-state value, as seen in fig. 4, is owed to the radial diffusion (section 3.6) resulting from our experiments' particular geometry).

One can infer from these considerations that it would seem reasonable to make the pumping target back surface effectively impenetrable for hydrogen, if one uses a "thin" target, i.e. if a substantial part of the

implanted hydrogen atoms is able to reach the opposite boundary before an appreciable re-emission begins on the side of implantation. Otherwise, one will not avail oneself of the surface passivation effect completely. The requirement of impenetrable outlet surface is met, for example, with a Nb–Cu "sandwich" (where the copper layer may also serve to conduct away the heat in the case of high power load on the pumping target).

Given $k_{r0}$ known (section 3.8), one can easily calculate the maximum concentration, $C$, and the maximum dose absorbed, $G = CL$, corresponding to the moment of time when the re-emission ratio[12], $j_0(t)/\xi j_i$, reaches the maximum tolerable amount, $\Psi$, for the case of "thin" membrane ($\xi$ here is the probability of implantation; for 10 keV protons, $\xi \approx 1$; for lower energies, with the reflection coefficient, $\delta$, known, we have $\xi = 1 - \delta$).

By the relation $\xi j_i \Psi = k_{r0} C^2$, we get

$$C = \sqrt{\xi j_i \Psi / k_{r0}}\,; \quad G = L\sqrt{\xi j_i \Psi / k_{r0}}\,. \tag{2}$$

The amounts of $C$ and $G$ computed with eq. (2) for the tolerable re-emission ratio $\Psi = 0.1$, and for $L = 0.1$ cm, $j = 1.5 \times 10^{15}$ $H^+/(cm^2\ s)$, are presented in fig. 7 alongside of $k_{r0}$.

Taking now a membrane of infinite thickness, it is clear that a characteristic depth, $\tilde{L}$, can be indicated beyond which the implanted hydrogen does not virtually penetrate before a fixed tolerable re-emission develops. An increase of the target thickness to amounts $L > \tilde{L}$ will not provide any significant increase of the maximum dose $G$. It is also clear that $\tilde{L}$ depends on the state of the surface: the stronger the evolution of implanted hydrogen is impeded, the larger $\tilde{L}$.

Using the evident equations,

$$\xi j_i \Psi = k_{r0} C^2(t); \quad C = \xi j_i t/\tilde{L}; \quad \tilde{L} = \sqrt{2Dt}\,; \quad G = C\tilde{L}, \tag{3}$$

we find $\tilde{L}$, $C$ and $G$ for a "thick" (with $L > \tilde{L}$) target:

$$\tilde{L} = 2D\sqrt{\frac{\Psi}{k_{r0}\xi j_i}}\,; \quad C = \sqrt{\frac{\xi j_i \Psi}{k_{r0}}}\,; \quad G = 2D\frac{\Psi}{k_{r0}}. \tag{4}$$

Substituting in eq. (4) $D = 5 \times 10^{-4} \exp(-10450\ \mathrm{J/mol\ H}/RT)$ $cm^2/s$ [5], and the $k_{r0}$ values from fig. 7, we get the amounts of $\tilde{L}$, $C$ and $G$ characteristic to our experimental conditions (viz. to our corpuscular flux and energy). These data are presented in fig. 8. Note that the expressions for the maximum dose, $G$, are essentially different in the cases of "thin" ($L \ll \tilde{L}$) and "thick" ($L > \tilde{L}$) target – cf. eqs. (2) and (4). In particular, the maximum dose for a thick target is much more strongly dependent on the state of the surface, since $G \propto k_{r0}^{-1}$, instead of $G \propto k_{r0}^{-1/2}$. This difference is clearly seen in fig. 8.

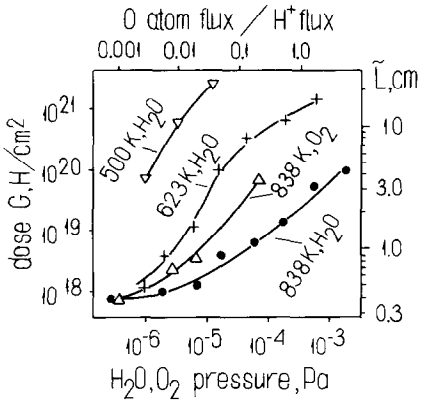



Fig. 8. Presented as functions of $H_2O$ and $O_2$ pressures are: $\tilde{L}$, the characteristic depth to that the absorbed hydrogen penetrates before the beginning of appreciable (with $\Psi = 0.1$) reemission, and $G$, the corresponding dose that a "thick" (with $L \geq \tilde{L}$) membrane is able to absorb. The equivalent flux of protons is $1.5 \times 10^{15}$ $H^+/(cm^2\ s)$, with the energy of 10 keV.

The presence of the chemically active gases effectively enhances the niobium capability to pump out fast hydrogen particles even in the case of thin target, as one can see from figs. 7 and 8. But one must take the targets of rather large thickness to use this effect to the full.

It is to be mentioned that the amounts of $G$, $C$, $\tilde{L}$ and $k_{r0}$ presented here refer to our specific parameters of experiment: $j_i = 1.5 \times 10^{15}$ $H^+/(cm^2\ s)$, with the energy per proton 10 keV. It may be hoped that bringing the energy down to hundreds or tens of eV will result in still more significant effects of active gases because of the decrease of ion sputtering. This trend has been demonstrated in our earlier experiments on hydrogen uptake and permeation in stainless steel in the presence of $H_2S$ [8].

### 4.2. Membrane pumping

We define $\Psi_\infty$, the ratio of re-emission for the steady-state superpermeability regime (i.e. with $j_{L\infty} \approx j_i$), as $\Psi_\infty = j_{0\infty}/\xi j_i$. On assuming $\xi \approx 1$, one gets

$$\Psi_\infty \approx j_{0\infty}/j_{L\infty}. \tag{5}$$

[12] We do not include reflected particles into the re-emission flux, $j_0$, in this treatment, since these particles retain their energy on the same scale [42], and they may be either pumped out at repeated collisions with the pumping surface, or returned to the working plasma.

If the re-emission occurs via the thermally controlled process, then

$$\Psi_\infty = \frac{k_{r0}C_{0\infty}^2}{k_{rL}C_{L\infty}^2}, \quad (6)$$

where $C_{0\infty}$ and $C_{L\infty}$ are the steady-state concentrations of dissolved hydrogen near the inlet and the outlet boundaries respectively. There are, practically, two possible cases of concentration distribution inside the membrane, except for a comparatively narrow range of temperature and of the values of $\alpha_L$ and $L$, transitional between the two extremes. One case is $C_{0\infty} \approx C_{L\infty}$ (the case of impeded evolution of dissolved hydrogen through the outlet boundary [1,2,8]). Then we have $\Psi_\infty \approx k_{r0}/k_{rL} = \alpha_0/\alpha_L$. The other case is $C_{0\infty} \gg C_{L\infty}$ (the case of relatively easy evolution through the outlet boundary arising with a sufficient increase of $\alpha_L$, $L$, $j_i$, and metal temperature). In this case

$$k_{r0}C_{0\infty}^2 = \Psi_\infty DC_{0\infty}/L, \; DC_{0\infty}/L = \xi j_i$$

holds true, whence $\Psi_\infty = L^2\xi j_i k_{r0}/D^2$. Combining the above we get

$$\Psi_\infty = \max\begin{cases} L^2\xi j_i k_{r0}/D^2 \\ k_{r0}/k_{rL} = \alpha_0/\alpha_L \end{cases}. \quad (7)$$

Thus $\Psi_\infty$ can never be smaller than $k_{r0}/k_{rL}$. If the tolerable $\Psi_\infty$ is larger than that, then the maximum membrane thickness and the maximum incident flux can be found that permit obtaining superpermeability with a pre-set re-emission ratio, $\Psi_\infty$:

$$L\sqrt{\xi j_i} \leqslant D\sqrt{\Psi_\infty/k_{r0}}, \text{ where } \Psi_\infty \geqslant k_{r0}/k_{rL}. \quad (8)$$

Inserting in eqs. (7) and (8) $k_r$ at different pressures of $H_2O$ and $O_2$ from Fig. 7, we obtain $\Psi_\infty$ and $L$ as functions of $H_2O$ and $O_2$ pressures. An example of estimating $L$ in this way is given by fig. 9. The presence of water vapour considerably increases the possible L's, but this seems to bear no practical importance, since even the smallest thicknesses of several millimetres indicated in fig. 9 are unlikely to cause any technical inconvenience in the applications. In other words, there is no thickness limitation for reaching superpermeability with niobium membranes.

The problem of compression is more stringent. Let $p$ be the pressure of molecular hydrogen by the outlet side of the membrane that is established on reaching the steady-state superpermeability regime. The upper estimate for the backstream flux of hydrogen permeating from the outlet to the inlet side may then be expressed [13] as $2Zp\alpha_0$. We put now the condition that this flux should not exceed a tolerable fraction, $\beta$, of the absorbed flux of fast particles: $2Zp\alpha_0 \leqslant \beta\xi j_i$. Defining the degree of compression, $g$, as $g = 2Zp/j_i$, we get for the maximum compression [2,13] $g_{max} = \beta\xi/\alpha_0 = (\beta\xi/\alpha_0^*)\exp(2\mathscr{E}_0/RT)$. Therefore, the attainable compression is to rise rapidly with the stronger passivation (i.e. with $\mathscr{E}_0$ rising and $\alpha_0^*$ diminishing [8,13]). To address our experimental data, it is more suitable to express $g_{max}$ via $k_{r0}$ by eq. (1):

$$g_{max} = \frac{\beta\xi Z}{S^2}\,\frac{1}{k_{r0}}.$$

Inserting $k_{r0}$ from fig. 6, with $S = 7\times10^{16}\exp(39700\ \text{J/mol H}/RT)$ atom H/(cm$^3$ Pa$^{1/2}$) [5], $Z = 1.1\times10^{19}$ molecule $H_2$/(cm$^2$ s Pa), and $\xi = 1$, we can now compute $g_{max}$ with a pre-set $\beta$. In fig. 10, $g_{max}$ is presented for $\beta = 0.1$. One sees that the degree of compression is larger at 838 K than at 623 K, whence it follows that $\alpha_0(838\text{ K}) < \alpha_0(623\text{ K})$. This is possible, if $\mathscr{E}_0$ and/or $\alpha_0^*$ are temperature dependent. The decrease of $\alpha_0$ with temperature may be related to a more rapid segregation of contaminants onto the surface at a higher temperature.

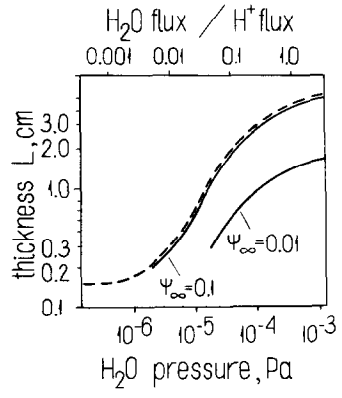



Fig. 9. The maximum thickness, $L$, of a niobium membrane that provides for the superpermeability characterized by the values of re-emission coefficient, $\Psi_\infty$, of 0.1 and 0.01 for the incident 10 keV proton flux of $1.5\times10^{15}$ $H^+$/(cm$^2$ s) at the membrane temperature of 838 K, and at varying the $H_2O$ pressure on the upstream side. The downstream surface is assumed to be either clean (solid curves), or protected by palladium overlayer (dashed curve).

[13] More accurately, the backstream flux equals $2Zp(1/\alpha_0 + 1/\alpha_L)^{-1}$ for the $C_{0\infty} \approx C_{L\infty}$ regime (e.g. ref. [8]). Considering that $\alpha_0 \leqslant \alpha_L$ (needed for superpermeability) one gets the estimate in the text. The factor 2 originates from that we calculate all the fluxes in the number of H-atoms.

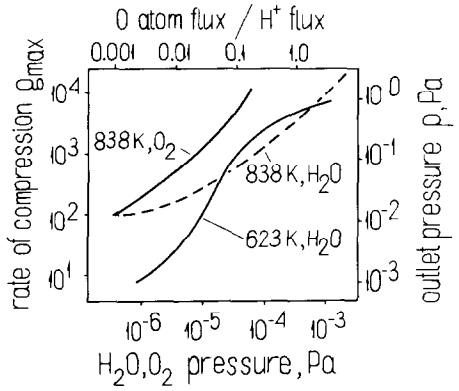



Fig. 10. Effects of $H_2O$ and $O_2$ on the attainable degree of compression, $g_{max}$, and on the outlet pressure of molecular hydrogen, $p$, with the condition that the backstream flow of hydrogen through the membrane does not exceed $0.1 j_i$. The equivalent incident flux, $j_i$, is $1.5 \times 10^{15}$ $H^+/(cm^2\ s)$, with the energy of 10 keV per proton.

Our data show that the presence of $H_2O$ or $O_2$ radically increases the attainable compression, though still much higher compression would be desired for practical purposes. This might be reached, in the case of Nb (V, Ta), by bringing down the rate of ion sputtering (lowering the energy), and by attempts at more effective surface passivation techniques, e.g. by $H_2S$ [8,16,19] or $C_2H_2$ [18] instead of $H_2O$ and $O_2$.

### *4.3. Some aspects of technical application*

Suggestions regarding the use in the fusion reactor technology of the membrane and sorptional pumping have been discussed in more detail elsewhere [8]. Several additional remarks may be relevant here.

Finding means of pumping the hydrogen particles of tens of eV energy may be particularly important, as such particles are expected to fall on the tokamak divertor plate, and also on the tokamak first wall. The data of our earlier experiments [16,28] give rise to hopes that the effective pumping by passivated niobium is possible of the ions having the tens of eV energy. The same experiments show that the thermal H-atoms are effectively pumped as well. The authors are engaged presently in systematic experiments with low energy ions [51] making use of the experimental equipment described in the present paper.

The data of the experiments with 10 keV per proton ions presented here may also have, in our opinion, a direct bearing upon the fusion technology. One of immediate engineering ideas is to use the membrane pumping in neutral beam injectors, both on earlier stages of acceleration, and in the reception target chamber, particularly, if the particles unabsorbed in the working plasma are retarded to recycle energy. The neutrals of several keV energy are also expected to fall on the linear in mirror devices, and their effective absorption is a matter of critical importance [31].

Admitting active gases to maintain or to restore the membrane surface passivation would not seem impermissible in beam injectors, since the membrane location may be relatively isolated from the working chamber of fusion reactor. The same pertains to the tokamak divertor pumping duct where a pumping membrane can be placed along the duct walls [52,53] to pump away the energetic particles after one or several bounces upon the wall. One might also think of evaporating a relatively non-volatile substance (e.g. carbon) hindering the release of implanted hydrogen onto the membrane inlet surface. In pulsed devices, the passivation of pumping surfaces can be conducted in the intervals between working pulses for the both options of sorptional and membrane pumping.

The pattern of behaviour qualitatively similar to that of niobium is to be expected from tantalum and vanadium, as all the three metals are known to exhibit a close similarity in the character of their interaction both with hydrogen and with non-metallic impurities typically present on the surface. Vanadium, though not being a refractory metal, presents the benefits of lower atomic number, lower induced radioactivity, and lower cost.

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